Evaluation of Noise-Robustness in Classical Instrument Classification

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Introduction

Background

- Instruments' timbral properties are central to how music is composed, performed, and perceived. As a result, instrument classification is an active research area (musicology, style analysis, audio editing and retrieval) [1, 2].
- Since 2015, convolutional neural networks have pushed clean-audio classification accuracy from ~90% to as high as ~99% [7].
- **The Gap:** model performance is usually measured on clean, isolated audio. It remains unclear which models fail when real-world noise is introduced, under what noise, and which instruments break down first.

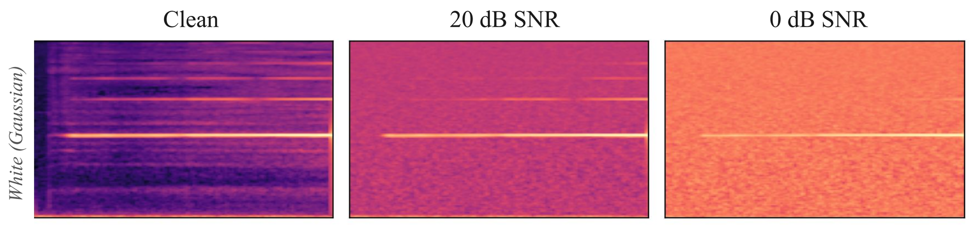

Fig.1 Flute A6 mel spectrogram; clean and under Gaussian noise

Objectives

- We run a controlled, head-to-head robustness comparison of six model families: handcrafted SVM; custom CNN and CRNN; and fine-tuned, pretrained MERT, AST, and PANNs.
- Trained on the same 12-instrument dataset, we hold the split and noise corpus identical across all models and add white (synthetic), human non-speech (ESC-50), and environmental (all 18 DEMAND environments) noise across an 8-level SNR sweep, measuring the change in classification performance.
- **Research Question:** How do instrument classification systems differ in their ability to retain classification performance under white, human non-speech, and environmental additive noise across a range of signal-to-noise ratios?

Methods

1 Source Recordings — Philharmonia Orchestra; 12 instruments; One labeled note / file →

2 Preprocessing — Filter articulation · tile silence; Resample · one 3 s window; Normalize loudness; → 8,374 retained recordings →

3 Grouped Split — Stratified by instrument, split by pitch 70/15/15; Train 5,861 · Val 1,258; Test 1,255 ↓

4 Train Models — Trained: SVM · CNN · CRNN; Fine-tuned: MERT · AST · PANNs →

5 Construct Noisy Test Set — White (Gaussian); Human non-speech (ESC-50); Environmental (DEMAND); 8 SNRs × 2 independent draws →

6 Evaluate / Score Models — Identical noisy mixtures per model; Macro-F1 retention across SNR

Fig.2 Methods pipeline

- 6 models were trained; 3 handcrafted, trained-from-scratch; 3 pretrained approaches
- Validation data used to select stopping decisions and hyperparameters, but not final model fitting
- Test data held out during model training; each final model tested once on clean test set

Dataset & Split

12 instruments	**8,374** windows	**3 s** @ 22.05 kHz
5,861 train	**1,258** validation	**1,255** test

Split is grouped by (instrument, note): every recording of a pitch lands in exactly one split, so near-duplicate takes cannot leak between train and test. A no-leak assertion runs on every dataset build.

Fig.3 Dataset and grouped split

Trained from scratch

No external pretraining

Model	Input to classifier	What's learned on this dataset
SVM	88 handcrafted audio features Timbre, spectrum, pitch, loudness	RBF decision boundary
CNN	128-band log-Mel spectrogram	Convolutional features + classifier
CRNN	128-band log-Mel spectrogram	Convolutional and recurrent Features plus classifier

Fig.4a Models trained from scratch

Pretrained backbones

Pretraining corpus listed beneath each model

Model	Input to classifier	What's learned on this dataset
MERT Music audio	24 kHz waveform 13 time-averaged layer outputs	Fine-tuned backbone Plus 12-class head
AST AudioSet	16 kHz waveform Official extractor to log-Mel	Transformer backbone Plus 12-class head
PANNs AudioSet	32 kHz waveform CNN14 computes 64-band log-Mel	CNN14 backbone Plus 12-class head

Fig.4b Models using pretrained backbones

Noise Construction

- Noise added to test set (1255 samples)
- SNR (Signal-to-noise ratio) is the ratio between the signal (audio source) compared to the noise. High SNR → faint noise; 0 dB SNR → noise is as loud as instrument

Eq. 1 · Power

$$P(x)=\frac{1}{N}\sum_{i=1}^{N}x_i^2$$

mean squared amplitude of a waveform

Eq. 2 · Seed

$$\text{seed}=\text{SHA256}(f|w|t|r)[:4]$$

fixes the noise draw (fingerprint, window, type, replicate)

Eq. 3 · Centering

$$n_c=n-\bar{n}$$

subtracts the mean to center the noise

Eq. 4 · Gain

$$\alpha=\sqrt{\frac{P(x)}{P(n_c)\cdot 10^{s/10}}}$$

multiplier that scales noise to the target SNR

Eq. 5 · Mixture

$$y=x+\alpha n_c$$

Eq. 6 · Check

$$\hat{s}=10\log_{10}\frac{P(x)}{P(y-x)}$$

- **white** — generated Gaussian noise — synthetic broadband control
- **human** — ESC-50 human non-speech (targets 20–29): 400 clips, 10 classes — clapping, coughing, laughing, footsteps …
- **environmental** — DEMAND: 18 real environments (domestic, nature, office, public, street, transport), one microphone per array

Every mixture is seeded by sha256(dataset fingerprint | window | noise type | replicate), so the 60,240-file corpus is bit-reproducible and identical for every model.

Results

Model	Clean macro-F1	AUC white	AUC human	AUC environ.
SVM	0.9788	0.2586	0.6490	0.5776
CNN	0.9708	0.3566	0.6526	0.5048
CRNN	0.9738	0.3302	0.6549	0.5152
MERT	0.9798	0.5734	0.7549	0.6857
PANNs	0.9885	0.4043	0.7779	0.7257
AST	**0.9908**	**0.6356**	**0.7906**	**0.7526**

Figure 5. Clean baseline and normalized robustness AUC — area under the retention-vs-SNR curve, dB-weighted (1.0 = no degradation across SNR). Bold marks the best value per column; model colours match Figure 6.

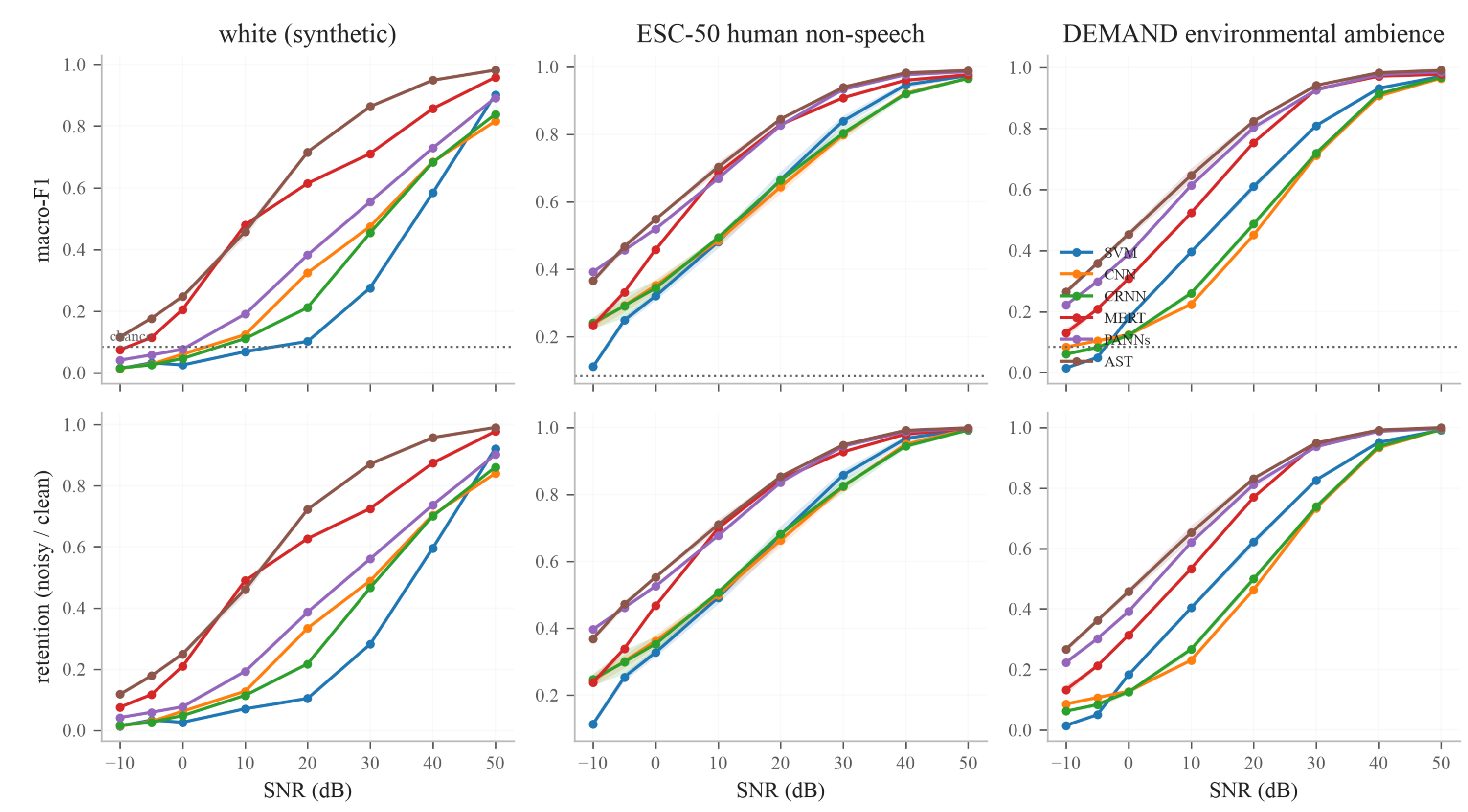

Figure 6. Macro-F1 (top) and retention relative to clean macro-F1 (bottom) versus SNR for all six models and three noise categories. Each point averages two noise realizations; shading shows their standard deviation.

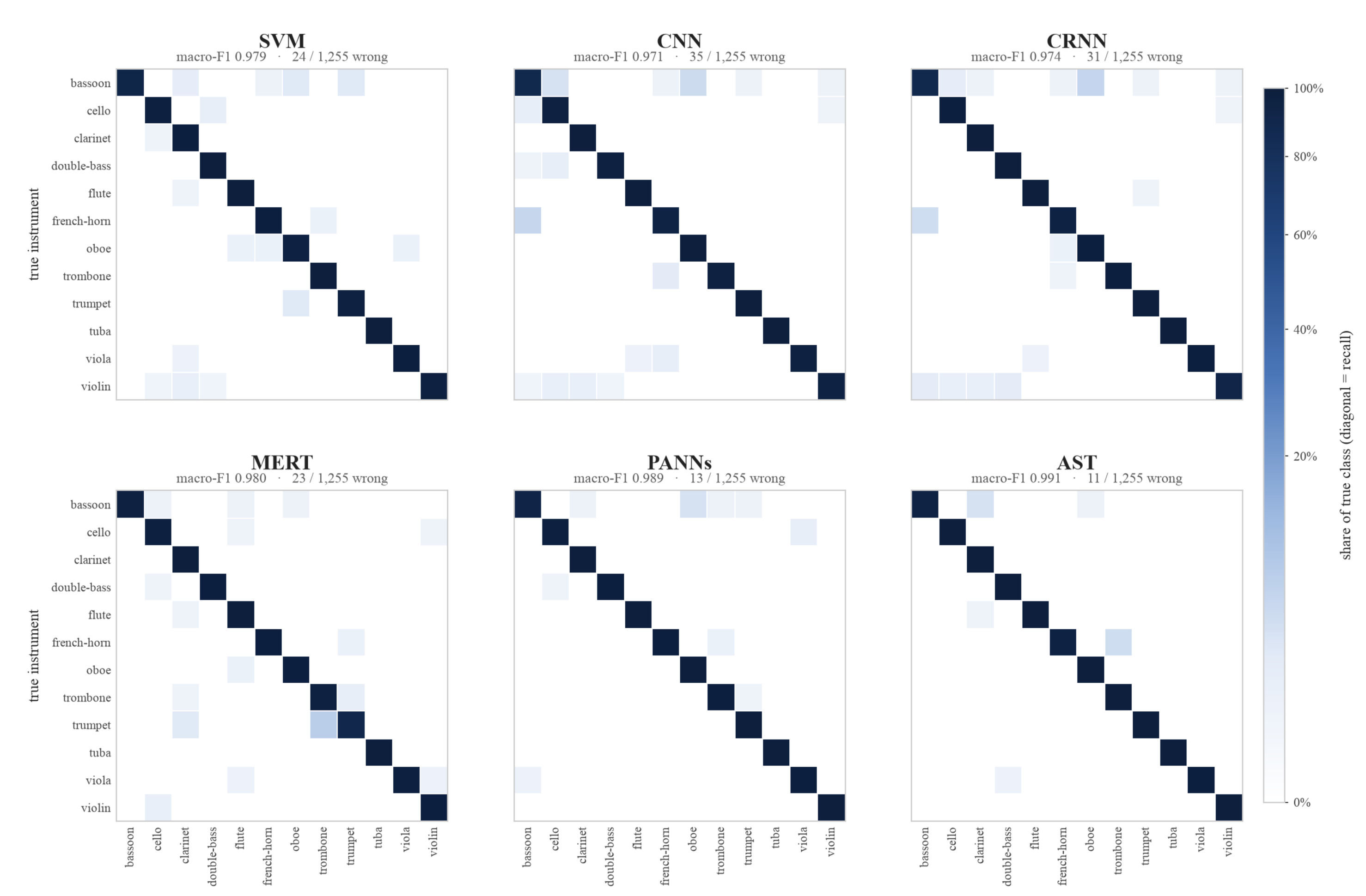

Figure 7. Clean-audio confusion for all six models, row-normalised: the diagonal is per-class recall, off-diagonal cells are the share of a true instrument sent elsewhere, on one colour scale across panels. Panel titles give clean macro-F1 and total misclassified windows.

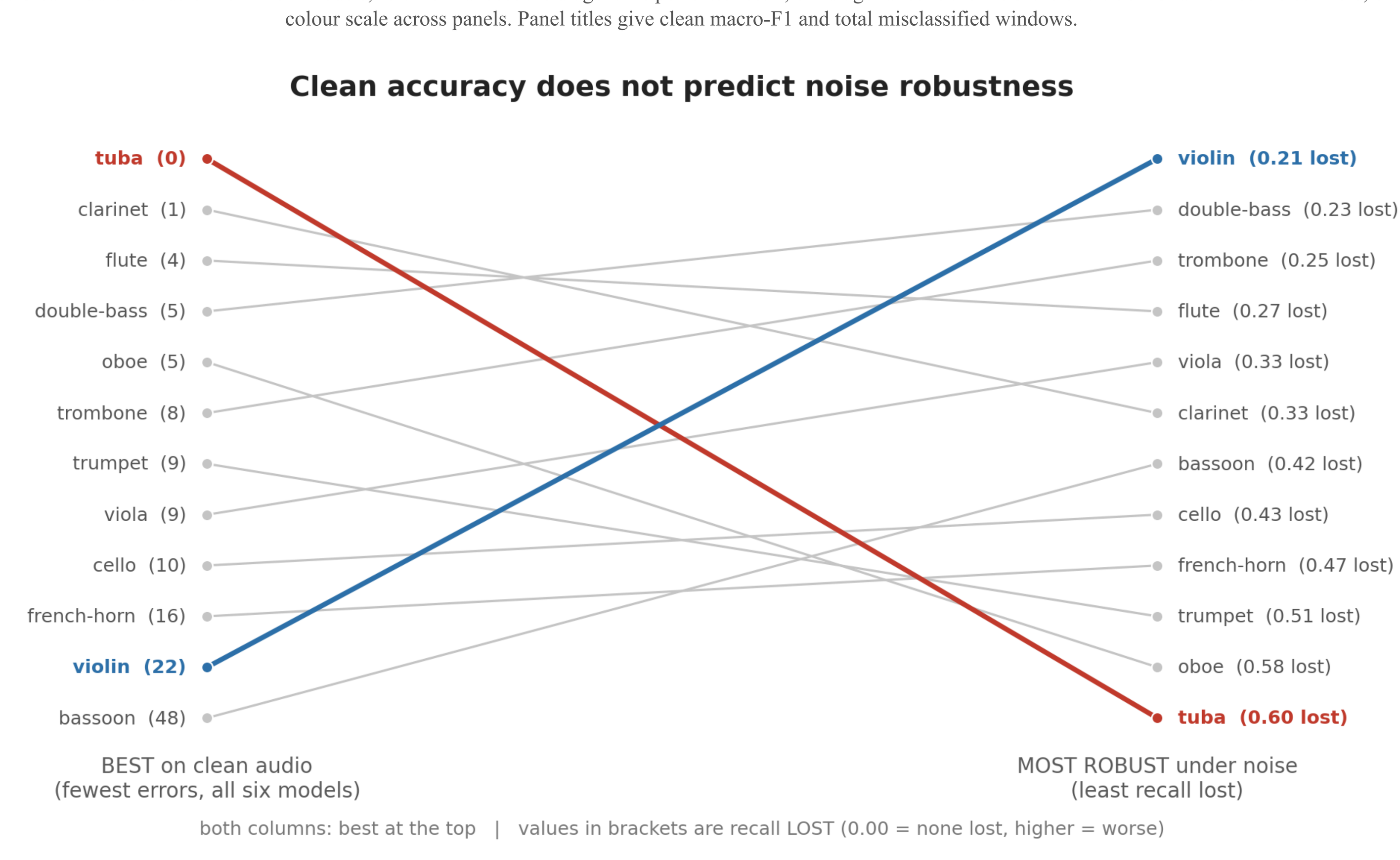

Figure 8. Instruments ranked by clean errors (left) and mean recall-loss AUC under noise (right), best at top. The rankings were unrelated (Spearman ρ = −0.18, p = 0.59). Tuba had no clean errors but the greatest recall loss (0.60); violin had 22 clean errors but the smallest loss (0.21).

- **Noise category mattered:** at 20 dB, SVM retained 10% under white noise and 68% under human non-speech noise.
- **Clean accuracy hid fragility:** tuba had no clean errors across the six models but the greatest mean recall-loss AUC under noise.
- **Failures were structured:** 7 of 18 distance-confusion tests remained significant after BH correction.
- **At 20 dB white noise,** CNN exceeded CRNN by 0.1126 macro-F1; both realization-specific bootstrap intervals excluded zero.

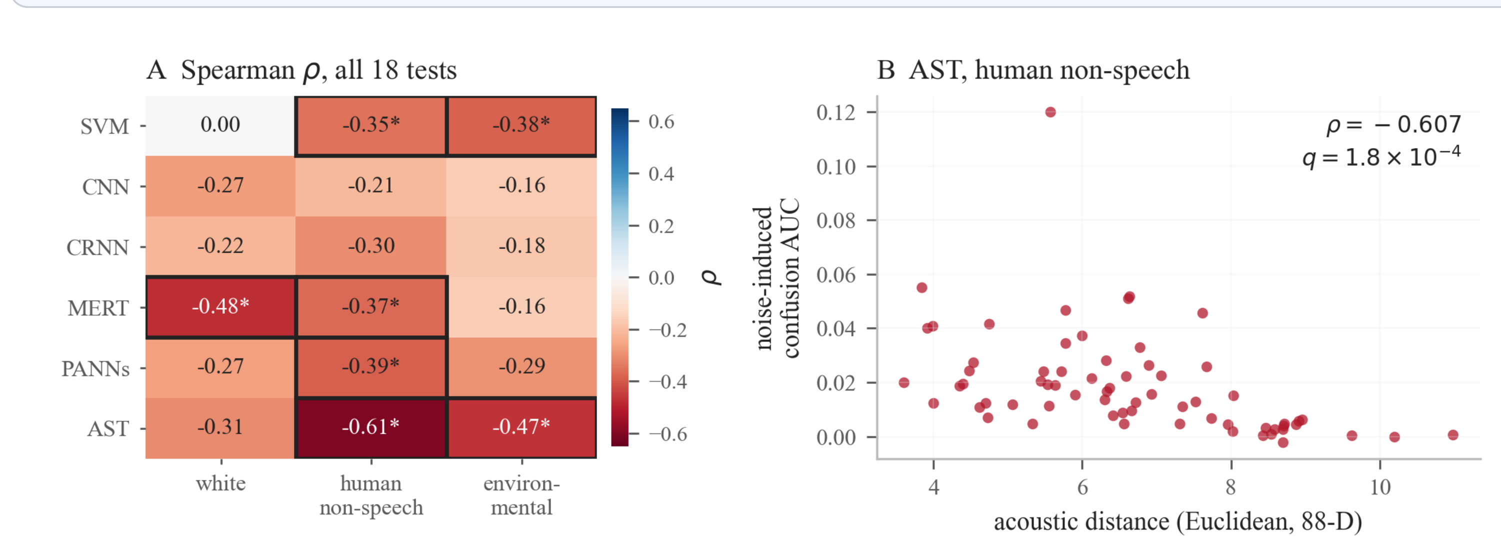

Figure 9. Acoustic distance and noise-induced confusion. Seven of 18 model-noise tests were significant after BH correction; AST under human non-speech noise had the strongest association.

Discussion

Evaluation Metrics

$$\text{Macro F1}=\frac{1}{12}\sum_{c=1}^{12}\text{F1}_c$$

Eq.7 Macro-F1 — primary evaluation metric; 12-instrument average

$$\text{Retention}=\frac{\text{Macro F1}_{noisy}}{\text{Macro F1}_{clean}}$$

Eq.8 Retention — standardized measure of robustness; how much clean score was retained after noise added

$$\text{AUC}=\frac{1}{s_{max}-s_{min}}\int_{s_{min}}^{s_{max}}\text{Retention}(s)\,ds$$

Eq.9 Area Under Robustness Curve — dB-weighted average retention across all eight noise levels

$$\text{Accuracy}=\frac{\#\,\text{correct}}{\#\,\text{examples}}$$

Eq.10 Accuracy — plain model correctness rate

Statistical Analysis

- Model differences were evaluated on identical test recordings; same clip with the same noise realizations
- Confidence intervals were estimated by resampling complete instrument-pitch groups
- Acoustic distance and noise-induced confusion were related using Spearman correlation. Significance assessed using 100,000 random instrument-label permutation tests, corrected using Benjamini–Hochberg correction to account for pure chance
- Paired comparisons used identical test windows and noise realizations. CNN exceeded CRNN by an average 0.1126 Macro-F1 over both realizations (0 & 1: 0.1122, 0.1130) under white noise at 20 dB SNR; 95% pitch-group bootstrap intervals r_0 = [0.0560, 0.1625] and r_1 = [0.0589, 0.1601] excluded zero
- Acoustically closer instrument pairs showed greater confusion; 7 of 18 model-noise tests are significant after Benjamini–Hochberg; strongest for AST under human non-speech (Spearman $\rho = -0.607$, permutation $p = 0.000010$, adjusted $q = 0.000180 < 0.05$).

Key Takeaways

- **Clean macro-F1 did not predict robustness.** All six clean scores spanned only 2.0 points, but white-noise retention AUC ranged from 0.259 (SVM) to 0.636 (AST).
- **AST had the highest observed robustness** under all 3 noise types (AUC 0.636 / 0.791 / 0.753). The weakest model varied: SVM under white and human non-speech, CNN under environmental noise.
- **Pretrained systems occupied the top three positions** under every noise type, but the study does not isolate pretraining from architecture, input, or sample rate.
- **Noise category mattered:** recorded noise was less damaging than white at matched nominal SNR. At 20 dB, AST retained 72.3% vs 85.3%; SVM retained 10.4% vs 68.0%.
- **Instrument failures were uneven:** tuba had the greatest overall recall-loss AUC. Acoustic distance was associated with confusion in 7 of 18 corrected tests, but the relationship is correlational.

Conclusion

- **Reproducible benchmark** spanning six instrument-classification systems
- **Evaluating robustness necessary;** clean accuracy alone is insufficient
- Noise category changes degradation at matched nominal SNR
- **Failures are structured.** Degradation concentrates in tuba, oboe, and trumpet. Confusion is associated with acoustic distance in 7 of 18 model/noise tests, most strongly for AST.
- **Shared, seed-reproducible noise** makes model comparison possible.

Limitations

- Only one dataset (Philharmonia), restricting timbre to one instrument, player, and/or recording setup
- Recordings are isolated notes, not polyphonic music where several instruments play at once
- Most audio clips were tiled to fill the 3-second standard, which confounds content with repetition
- Nominal full-band SNR and model-specific frontends can change effective masking; cross-model differences describe complete systems
- SVM, AST, MERT and PANNs are single-seed runs, while CNN and CRNN are multi-seed spread, so small differences are not treated as effects

Future Work

- Polyphonic and multi-dataset audio
- Independently recorded real-world noise, more real acoustic issues
- More noise realizations to increase robustness against random draw
- Noise-aware training to further noise robustness

Reproducibility

- All per-window predictions, per-condition metrics, and the failure analysis are committed to the repository
- Every figure and table on this poster regenerates from one script each (scripts/*.py) against those files
- The noise corpus rebuilds bit-identically from the sealed dataset fingerprint

References

Works Cited

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